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Short-arc tracklet association for geostationary objects

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Abstract

Measurement association and initial orbit determination is a fundamental task when building up a database of space objects. This paper proposes an efficient and robust method to determine the orbit using the available information of two tracklets, i.e. their line-of-sights and their derivatives. The approach works with a boundary-value formulation to represent hypothesized orbital states and uses an optimization scheme to find the best fitting orbits. The method is assessed and compared to an initial-value formulation using a measurement set taken by the Zimmerwald Small Aperture Robotic Telescope of the Astronomical Institute at the University of Bern. False associations of closely spaced objects on similar orbits cannot be completely eliminated due to the short duration of the measurement arcs. However, the presented approach uses the available information optimally and the overall association performance and robustness is very promising. The boundary-value optimization takes only around 2% of computational time when compared to optimization approaches using an initial-value formulation. The full potential of the method in terms of run-time is additionally illustrated by comparing it to other published association methods.
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Keywords: Space situational awareness; Initial orbit determination; Tracklet association; Too-short-arcs; Admissible region

1. Introduction

In order to avoid accident-prone proximities of active satellites and uncontrolled space objects or eventually to remove space debris, space object catalogs must be maintained. The geostationary orbit is of special importance as it is intensively used for communication (e.g. Astra and Eutelsat satellites), navigation (e.g. the European Geostationary Navigation Overlay Service) and weather monitoring (e.g. Meteosat satellites). Additionally, it is special because of its far distance from the surface of the Earth. Due to the limited range capabilities of radar antennas, it is usually observed with optical telescopes. Considering the limited amount of telescopes trying to cover the complete orbital region, each object can only be tracked for a limited duration. The resulting short observation arcs, called tracklets, only contain incomplete state information and are therefore either associated to already cataloged objects (e.g. with the procedure described in Früh et al., 2009) or tested pairwise with other uncorrelated observations. The latter problem is approached in this work, i.e., two tracklets are examined whether they originate from the same object or not and, if they do, the common orbit solution is determined. This measurement association is a fundamental task during the catalog build-up phase for the initial location of space objects but also later for the relocation of lost ones.

1.1. Observations

Each measurement arc contains a series of right ascension (α) and declination (δ) values as measured by a

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topocentric observer. The information of the series, which is exploited for the association, is the line-of-sight $\mathbf{u}$ and its derivative $\dot{\mathbf{u}}$. It can also be represented by the angles and angular rates, commonly known as the attributable vector (c.f. Milani et al., 2004)

$$\mathbf{a} = (\alpha, \dot{\alpha}, \delta, \dot{\delta})^{\top}. \tag{1}$$

The use of the derivatives is only justified as long as measurement errors are small and the tracklet length is long enough for consistent catalog maintenance.

1.2. Problem formulation

The orbital motion of an object is described by six first order ordinary differential equations, where the vector form of the system is shown below

$$\dot{\mathbf{y}}(t) = \mathbf{f}(t, \mathbf{y}(t)) \quad \text{where} \quad \mathbf{y}(t) = \begin{pmatrix} \mathbf{r}(t) \\ \dot{\mathbf{r}}(t) \end{pmatrix}. \tag{2}$$

The state of the system is uniquely defined by six constraints, i.e. an orbit solution is either defined by an initial value

$$\mathbf{y}(t_1) = \begin{pmatrix} \mathbf{r}_1 \\ \dot{\mathbf{r}}_1 \end{pmatrix} \tag{3}$$

or by two boundary values

$$\mathbf{y}_{1:3}(t_1) = \mathbf{r}_1 \quad \text{and} \quad \mathbf{y}_{1:3}(t_2) = \mathbf{r}_2, \tag{4}$$

where in the first case (Eq. (3)) the position and velocity at one epoch must be provided and in the latter case (Eq. (4)) the positions at two observation epochs are needed. The subscript denotes the constrained elements of the state vector. The different components of the state vector can be described in terms of the observed variables. The geocentric position

$$\mathbf{r}(\rho) = \mathbf{r}_s + \rho\,\mathbf{u} \tag{5}$$

is dependent on the line-of-sight information, the sensor position $\mathbf{r}_s$, and the unobserved range ρ. The velocity

$$\dot{\mathbf{r}}(\rho, \dot{\rho}) = \dot{\mathbf{r}}_s + \rho\,\dot{\mathbf{u}} + \dot{\rho}\,\mathbf{u}, \tag{6}$$

additionally requires knowledge on the line-of-sight derivative, the sensor velocity, and the unobserved range-rate $\dot{\rho}$.

Therefore, one tracklet constrains the state of an object in four degrees of freedom with two unknown parameters $(\rho, \dot{\rho})$. Thus, to uniquely determine a state of an object, at least two observation arcs are required. Hence, an overdetermined system with eight known and four free parameters, namely the range and range-rate at both observation epochs is obtained. The tracklet association problem is illustrated in Fig. 1.

To date, various methods have been developed to approach the tracklet association problem. All methods hypothesize some of the unknown free parameters to obtain an orbit estimate. Then, these hypotheses are tested and either accepted or rejected. However, the methods differ in the way the orbit is represented, i.e. using the initial value or the boundary value formulation. Both formulations come with certain drawbacks and benefits, which will be discussed in Section 6.3. Table 1 gives an overview of current approaches using the different formulations and their relation to each other. They are recapitulated in the following.

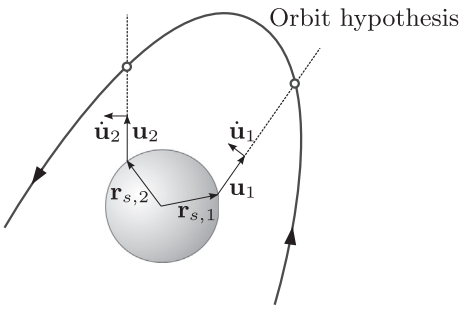


Fig. 1. Illustration of the two tracklet association problem: Shown are two line-of-sight vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ and their derivatives at different epochs and with different station coordinates $\mathbf{r}_{s,1}$ and $\mathbf{r}_{s,2}$. Furthermore, an orbit hypothesis connecting both observations is shown.

2. Initial-value method

When augmenting the tracklet information from the first observation epoch with the two free parameters $(\rho_1, \dot{\rho}_1)$, a full state is defined

$$\hat{\mathbf{y}}(t_1) = \begin{pmatrix} \mathbf{r}(\rho_1) \\ \dot{\mathbf{r}}(\rho_1, \dot{\rho}_1) \end{pmatrix}, \tag{7}$$

which can be used as an initial value to numerically or analytically integrate the equation of motion (2) to the second epoch. Consequently, the solution $\hat{\mathbf{y}}(t_2)$ can be computed. The hat over the state variable indicates that the orbital solution is a hypothesis and not necessarily the true one. The observation arc at the second epoch t_2 serves as a discriminator to decide whether a hypothesis is accepted or rejected.

Three different methods have been published which use this formulation. The solution space of the problem is limited to a so called admissible region in all these methods. Milani et al. (2004) and Tommei et al. (2007) suggest to restrict the range and range-rate space by allowing only solutions that orbit the Earth on stable orbits, i.e. the energy of the candidate orbits must be negative and the distance between the object and the observer must be above a certain limit. Maruskin et al. (2009) additionally defined bounds on apoapsis and periapsis radii. The latter assures that the candidate solution does not de-orbit within the next revolutions. If only a specific orbital region is of interest, the extent of the region can be furthermore reduced by specifying semi-major axis or eccentricity bounds (DeMars and Jah, 2013; DeMars et al., 2012).

2.1. Regular grid testing

DeMars et al. (2012) sample the admissible region of the first tracklet with multiple hypotheses on a regular grid to

Table 1
Overview of the current tracklet association approaches.

Orbit representation	Parameters		Approaches	
Initial value $\mathbf{r}_1$ and $\dot{\mathbf{r}}_1$	State	$(\alpha_1, \dot{\alpha}_1, \delta_1, \dot{\delta}_1)$	Regular grid testing	DeMars et al., 2012
	Hypotheses	$(\rho_1, \dot{\rho}_1)$	Hyperplane intersection	Fujimoto et al. (2013)
	Discriminator	$(\alpha_2, \dot{\alpha}_2, \delta_2, \dot{\delta}_2)$	Optimization	Siminski et al. (2013a,b)
Boundary value $\mathbf{r}_1$ and $\mathbf{r}_2$	State	$(\alpha_1, \delta_1, \alpha_2, \delta_2)$	Regular grid testing	Schumacher et al. (2013)
	Hypotheses	(ρ_1, ρ_2)	Optimization	Presented approach
	Discriminator	$(\dot{\alpha}_1, \dot{\delta}_1, \dot{\alpha}_2, \dot{\delta}_2)$		

initialize a filter. Each $(\rho_1, \dot{\rho}_1)$ – hypothesis is represented by a Gaussian, effectively a state $\hat{\mathbf{y}}(t_1)$ and covariance. This set of hypotheses statistically represents the complete admissible region and is then updated given the following measurements. Hence, modeled measurements $\hat{\mathbf{a}}_2$ are computed from the propagated state hypotheses at the next epoch $\hat{\mathbf{y}}(t_2)$ and tested with the actual other observation $\mathbf{a}_2$. If the association probability falls below a predefined threshold, the component is removed from the distribution. The surviving hypotheses create candidate objects, which are affirmed and refined with further observations. Gadaleta et al. (2012) use the same principle for association tests only. They also assess the association performance for different probability measures.

2.2. Hyperplane intersection

Fujimoto et al. (2013) find the common solution by computing the overlap of two reduced solutions spaces. Each admissible region is discretized with $(\rho_1, \dot{\rho}_1)$ - and $(\rho_2, \dot{\rho}_2)$ – hypotheses, which are represented by uniform probability density functions. All hypotheses are then propagated to a common epoch and compared to each other in some orbital elements space, which effectively is an intersection of two hyperplanes. The intersecting hypotheses are affirmed or rejected using a least-squares fit.

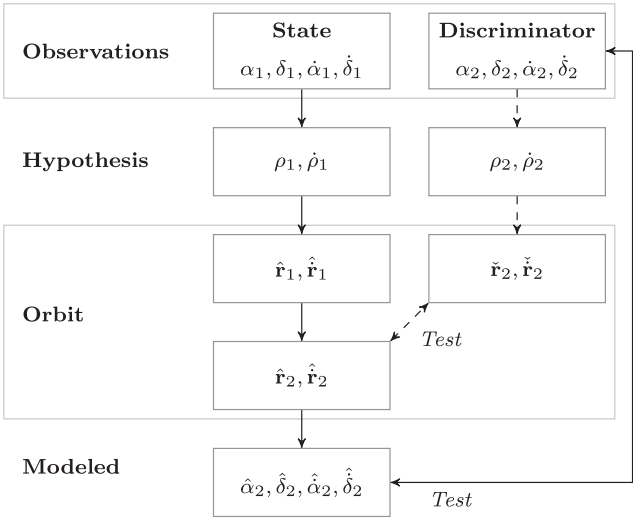


Fig. 2. Process flow of the initial-value methods. The dashed path denotes the testing approach using hyperplane intersection.

Both methods are illustrated in Fig. 2, where the dashed lines represent the process flow from Fujimoto et al. (2013) and the solid lines the one from DeMars and Jah (2013).

2.3. Optimization

In a previous work, a method was introduced which searches through the free parameter space rather than evaluates all possible points on a regular grid (Siminski et al., 2013a). As in the approach by DeMars and Jah (2013) and Gadaleta et al. (2012) the hypothesis states are defined at one epoch only. However, the association probability is not evaluated for all points on the grid but computational optimization schemes are exploited instead to identify whether two observations originate from the same object or not.

3. Details on the initial-value optimization

A loss function is minimized which describes the difference between the actual second observation $\mathbf{a}_2$ and the modeled value $\hat{\mathbf{a}}_2$. As the measurements are affected by noise, the difference in the loss function is scaled with its uncertainty. The loss function is then given by

$$L(\mathbf{p}) = (\mathbf{a}_2 - \hat{\mathbf{a}}_2)^\top (\mathbf{C}_{\hat{\mathbf{a}}_2} + \mathbf{C}_{\mathbf{a}_2})(\mathbf{a}_2 - \hat{\mathbf{a}}_2), \tag{8}$$

where the range and range-rate hypotheses are combined in $\mathbf{p} = (\rho_1, \dot{\rho}_1)$ for convenience and

$$\hat{\mathbf{a}}_2(\mathbf{a}_1, \mathbf{p}) = (\hat{\alpha}_2, \hat{\dot{\alpha}}_2, \hat{\delta}_2, \hat{\dot{\delta}}_2)^\top, \tag{9}$$

is the propagated modeled observation obtained from $\hat{\mathbf{y}}(t_2)$.

The covariance matrix $\mathbf{C}_{\mathbf{a}_2}$ describes the uncertainty of the second measurement and $\mathbf{C}_{\hat{\mathbf{a}}_2}$ is the propagated covariance matrix of the initial measurement

$$\mathbf{C}_{\hat{\mathbf{a}}_2} = \left(\frac{\partial \hat{\mathbf{a}}_2(\mathbf{a}_1, \mathbf{p})}{\partial \mathbf{a}_1}\right)^\top \mathbf{C}_{\mathbf{a}_1} \left(\frac{\partial \hat{\mathbf{a}}_2(\mathbf{a}_1, \mathbf{p})}{\partial \mathbf{a}_1}\right). \tag{10}$$

To justify the use of the loss function, it can be interpreted as a measure for the association probability with multiple hypotheses (c.f. Bar-Shalom et al., 2004).

Given any hypothetical state $\mathbf{p}$, the probability of both tracklets originating from the same object can be computed using Bayes' theorem

$$P(\mathbf{p}|\mathbf{a}_2, \mathbf{a}_1) = \frac{P(\mathbf{a}_2|\mathbf{p}, \mathbf{a}_1) P(\mathbf{p}, |\mathbf{a}_1)}{P(\mathbf{a}_2|\mathbf{a}_1)}, \tag{11}$$

where the likelihood, assuming that it is normally distributed, is given by

$$P(\mathbf{a}_2|\mathbf{p},\mathbf{a}_1) = \mathcal{N}(\mathbf{a}_2;\hat{\mathbf{a}}_2,\mathbf{C}_{\hat{\mathbf{a}}_2}+\mathbf{C}_{\mathbf{a}_2}), \tag{12}$$

The Gaussian probability density function is calculated with

$$\mathcal{N}(\mathbf{x};\mu,\mathbf{C}) = |2\pi\mathbf{C}|^{-\frac{1}{2}}\exp\{-\frac{1}{2}(\mathbf{x}-\mu)^{\top}\mathbf{C}^{-1}(\mathbf{x}-\mu)\}, \tag{13}$$

where $\mathbf{x}$ is the random variable, μ the mean, and $\mathbf{C}$ the respective covariance matrix.

As the evidence is independent of the free parameters and the prior is constant for all possible orbits, only the likelihood is assessed. The hypothesis with the largest likelihood is accordingly the best fitting orbit to the second tracklet. Alternatively, the free parameters can be found that minimize the exponent of the likelihood, i.e. the above shown loss function (Eq. 8). That comes with the advantage that the result is distributed according to the χ^2-distribution and can be gated using predefined thresholds (DeMars et al., 2012).

The topography of the loss function for the free parameters $\mathbf{p} = (\rho_1, \dot{\rho}_1)$ is illustrated in Fig. 3, where the same geostationary object is re-observed after 3.75 days. The semi-major axis and orbital period stay constant on the dashed lines. The numbers in brackets denote the range of revolutions k (in orbital periods) in the enclosed regions. The true solution is in the range $k \in [3.5, 4]$.

The topography of the loss function has been analyzed in Siminski et al. (2013a) to assess the suitability for computational optimization methods. Several hills and valleys can be observed and, consequently, several local minima inside the valleys can be found that are potential candidates for a common orbit solution of the two measurement arcs. Fujimoto and Scheeres (2012) explain why these multiple revolution solutions can lead to false associations. This results in a multi-modal optimization problem and makes it difficult to find a global minimum within the admissible region. Typically, optimization methods start with an initial value and then improve the solution by gradually moving towards the next minimum using e.g. a slope approximation. Therefore, when using these methods, the location of the resulting minimum is largely dependent on the initial value, e.g. it is likely to find the local minimum in the same valley from which the search is started.

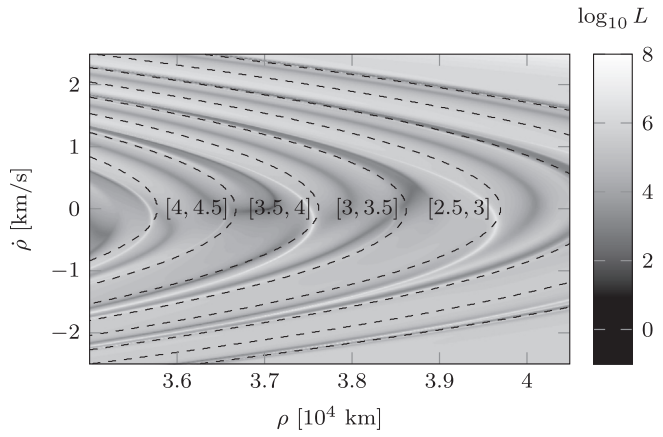


Fig. 3. Loss function for initial-value method with range and range-rate as the free parameters. The same geostationary object at the longitude $\lambda = 10°$ has been re-observed after 3.75 days. The observing sensor is located at the same longitude and latitude $\phi = 45°$.

For that reason, the individual valleys must be identified beforehand. They are roughly separated by allowing only a specific range of orbital revolutions k between the observation epochs. However, multiple valleys inside the bounds can still be observed and this separation can lead to very narrow regions as observed in Fig. 3, which is disadvantageous for iterative search algorithms as these might step outside the feasible area too easily. An appropriate alternative parameter system has been therefore proposed in Siminski et al. (2013b) to account for the difficult topography. The semi-major axis a_1 and a relative range $\tilde{\rho}_1$ at the first epoch are used as the free parameters. The derivation and description of the new parameters can be obtained from Siminski et al. (2013b).

The topography of the loss function for the new set of parameters is illustrated in the following Fig. 4. It shows a more suitable pattern for optimization algorithms. The representation allows one to split the loss function into rectangles with the same number of orbital revolutions. Even though most valleys have been successfully separated from each other, several hills and valleys inside an orbital revolutions range can still be seen. Future research must find ways to identify the different valleys inside such a range. If the location or extent of all valleys are known, each one can be individually minimized using a gradient-based optimizer. For now, the multi-modal optimization problem still needs to be solved.

4. Boundary-value method

Instead of parametrizing the problem at the initial epoch only, more symmetric approaches have been developed that use the boundary-value formulation to define a state

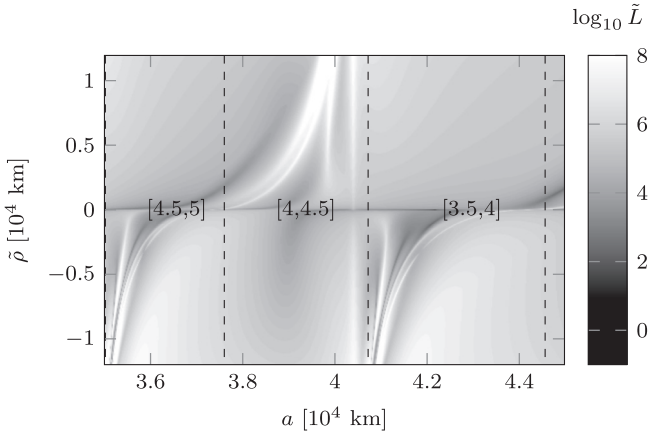


Fig. 4. Loss function for initial-value method with semi-major axis and relative range as the free parameters. The same observations are used as in the previous Fig. 3.

hypothesis. The line-of-sight information at both observation epochs

$$\mathbf{z} = (\alpha_1, \delta_1, \alpha_2, \delta_2)^\top \tag{14}$$

is used to constrain the state of the tested hypotheses. A complete orbital state is then defined by augmenting the line-of-sight information with unknown range values at both epochs (ρ_1, ρ_2)

$$\hat{\mathbf{y}}_{1:3}(t_1) = \mathbf{r}(\rho_1) \quad \text{and} \quad \hat{\mathbf{y}}_{1:3}(t_2) = \mathbf{r}(\rho_2). \tag{15}$$

4.1. Regular grid testing

Schumacher et al. (2013) propose to use only the line-of-sight information if the angular rates are not available within a sufficient accuracy. Therefore, they sample a feasible range space for all measurements. Then, each range hypothesis of one measurement is combined with the range hypotheses of the other measurements. The solution to the Lambert's problem is used to determine an orbit for each (ρ_1, ρ_2) - hypothesis. Then, the generated orbit hypotheses are affirmed and refined with further measurements.

Ansalone and Curti (2013) present a method for an observer in a low Earth orbit, which uses the same idea. They sample the feasible range space on a coarse grid and test each hypothesis with the individually measured angles of the two tracklets. The fitness of each hypothesis is then iteratively improved using a genetic algorithm. Similarly, Schneider (2013) performes so called statistical ranging to determine orbit candidate solutions where he characterizes the error distributions via Markov Chain Monte Carlo. The probability distributions are then used to group tracks originating from the same objects.

4.2. Optimization

This work introduces a method, where, again, an optimization scheme will be used to identify tracklets of common objects. The respective range values at both epochs are again chosen as free parameters to create candidate solutions. But, instead of testing the orbit hypotheses with further measurements or the individually measured angles, the angular rates

$$\dot{\mathbf{z}} = (\dot{\alpha}_1, \dot{\delta}_1, \dot{\alpha}_2, \dot{\delta}_2)^\top \tag{16}$$

are taken into account to decide whether a solution is suitable or not. These parameters are then used to define a loss function which is minimized to find the best fitting orbit to both observations.

Fig. 5 illustrates the process flow of the boundary-value methods. The dashed line represents the presented approach, while the method proposed by Schumacher et al. (2013) uses additional measurements (see Fig. 6).

5. Details on the boundary-value optimization

The two range hypotheses are combined for convenience in one hypothesis variable $\mathbf{p} = (\rho_1, \rho_2)$. When a

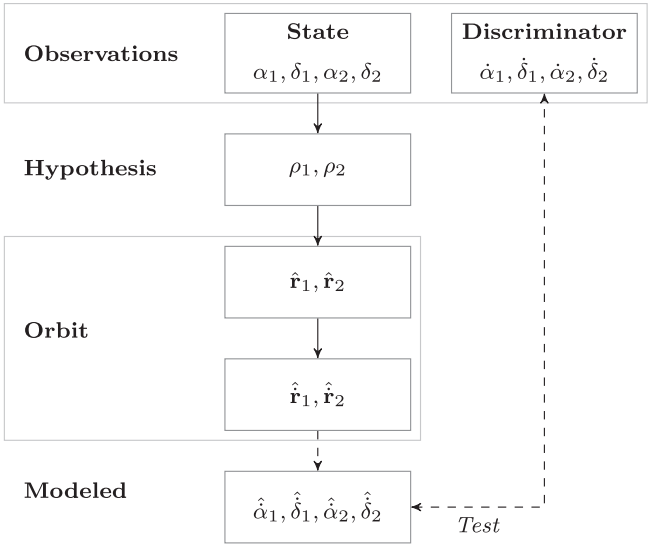


Fig. 5. Process flow of the boundary-value methods. The dashed line denotes the new testing metric.

$\mathbf{p}$ - hypothesis is given, the geocentric position vectors $\hat{\mathbf{r}}_1$ and $\hat{\mathbf{r}}_2$ are obtained using the line-of-sight information $\mathbf{z}$ and the sensor positions at the respective epochs. This forms a special case of an orbital boundary value problem, commonly known as Lambert's problem, where the boundaries for the equation of motion (2) are constrained by two position vectors at epochs t_1 and t_2.

Several researchers solved it using geometrical considerations for non-perturbed two-body dynamics. Practical implementations can be obtained from e.g. Gooding (1990) or Battin (1999). As the problem allows multiple solutions depending on the transfer path and the number of orbital periods, the number of full and half revolutions k needs to be beforehand specified. If the fractional part of k is larger than 0.5, the long transfer path around the orbital ellipse is chosen. Otherwise, the solution is computed for the short way. The solution to the problem, effectively a set of orbital elements, is then used to determine the geocentric velocities $\dot{\hat{\mathbf{r}}}_1$ and $\dot{\hat{\mathbf{r}}}_2$ at both observation epochs. From these, the modeled topocentric rates

$$\dot{\hat{\mathbf{z}}}(\mathbf{z}, \mathbf{p}, k) = (\dot{\hat{\alpha}}_1, \dot{\hat{\delta}}_1, \dot{\hat{\alpha}}_2, \dot{\hat{\delta}}_2)^\top \tag{17}$$

are obtained.

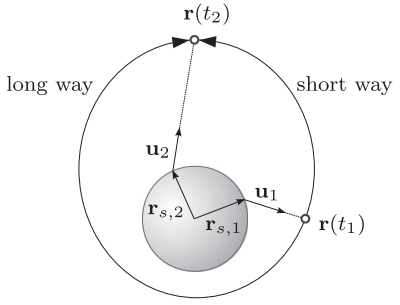


Fig. 6. Illustration of Lambert's problem. The two possible transfer path directions are demonstrated.

As discussed before (Section 4.2), the four remaining observables $\dot{\mathbf{z}}$ are utilized in a loss function as a discriminator

$$L(\mathbf{p},k) = (\dot{\mathbf{z}} - \hat{\dot{\mathbf{z}}})^{\top}(\mathbf{C}_{\dot{\mathbf{z}}} + \mathbf{C}_{\hat{\dot{\mathbf{z}}}})(\dot{\mathbf{z}} - \hat{\dot{\mathbf{z}}}), \tag{18}$$

where the parameters $\mathbf{p}$ and the number of revolutions and the transfer path, i.e. the variable k, are all unknown.

The loss function is effectively the difference between the measured and the modeled angular rates scaled by its uncertainty. The covariance matrix $\mathbf{C}_{\dot{\mathbf{z}}}$ describes the uncertainties in the angular rates from (16) and $\mathbf{C}_{\hat{\dot{\mathbf{z}}}}$ represents the effect of the angular uncertainties from (14) on the uncertainty of the modeled angular rates $\hat{\dot{\mathbf{z}}}$.

The covariance matrix $\mathbf{C}_{\mathbf{z}}$ for the angles must be transformed using

$$\mathbf{C}_{\hat{\dot{\mathbf{z}}}} = \left(\frac{\partial \hat{\dot{\mathbf{z}}}(\mathbf{z},\mathbf{p},k)}{\partial \mathbf{z}}\right)^{\top} \mathbf{C}_{\mathbf{z}} \left(\frac{\partial \hat{\dot{\mathbf{z}}}(\mathbf{z},\mathbf{p},k)}{\partial \mathbf{z}}\right). \tag{19}$$

The use of the loss function is justified similarly as in the case of the loss function in Eq. (8). Following the derivation in Section 3, the probability for a $\mathbf{p}$ – hypothesis and a fixed k, given the two measurement arcs, is assessed using Bayes' theorem

$$P(\mathbf{p},k|\dot{\mathbf{z}},\mathbf{z}) = \frac{P(\dot{\mathbf{z}}|\mathbf{p},k,\mathbf{z})P(\mathbf{p},k|\mathbf{z})}{P(\dot{\mathbf{z}}|\mathbf{z})}, \tag{20}$$

where the likelihood is given by

$$P(\dot{\mathbf{z}}|\mathbf{p},k,\mathbf{z}) = \mathcal{N}\left(\dot{\mathbf{z}};\hat{\dot{\mathbf{z}}},\mathbf{C}_{\hat{\dot{\mathbf{z}}}} + \mathbf{C}_{\dot{\mathbf{z}}}\right), \tag{21}$$

assuming that latter is normally distributed. The function $\mathcal{N}$ is defined in Eq. (13). Analogously to Section 3, only the likelihood is assessed for the association probability. The hypothesis with the largest likelihood is accordingly the best fitting orbit to the measured angular rates. Then, again, the free parameters can be found that minimize the exponent of the likelihood in order to allow gating. Thus, if $\min\{L(\mathbf{p},k)\}$ falls below the previously defined threshold, the hypothesis is accepted and both tracklets are associated to each other.

The topography of the loss function is again evaluated to study the suitability for computational optimization schemes. Fig. 7 shows the loss function for the same example case as for the initial-value method. Three individual discrete cases of the loss function defined by the allowed k interval have been created. In contrast to the initial-value method, each discrete case of the loss function contains only one local and therefore also global minimum. The global minimum for the overall problem is then obtained by identifying the value of k that yields the smallest loss-function minimum.

5.1. Admissible region

As in the case of the initial-value formulation, the solution space for $\mathbf{p}$ - hypotheses is constrained to an admissible

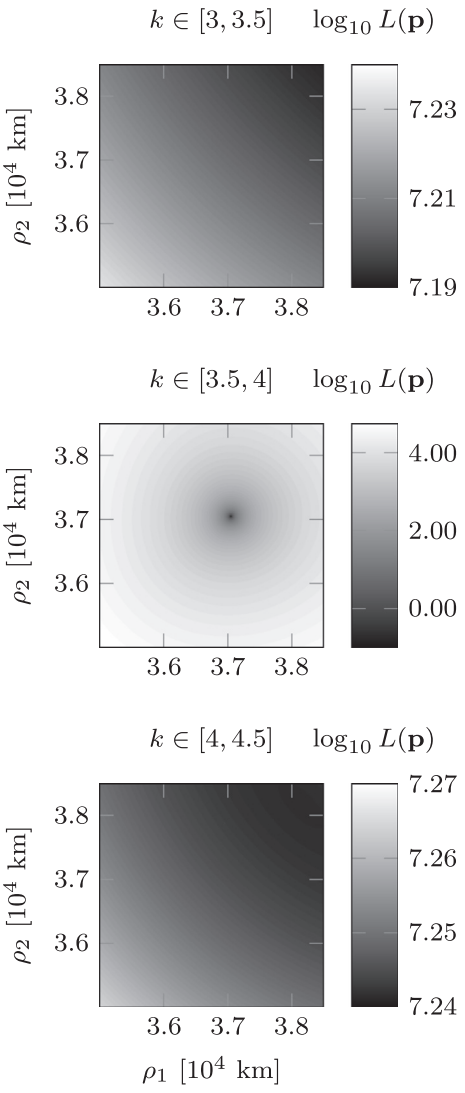


Fig. 7. Loss function for boundary-value method. The same observations were used as in the previous Figs. 3 and 4. The loss function is evaluated for different ranges of allowed orbital periods k.

region $\mathcal{C}$. In principle, the complete information of the tracklets, i.e. line-of-sight and derivative, can be used to determine the extent of the feasible region. A methodology to obtain such an admissible region $\mathcal{C}(\mathbf{z},\dot{\mathbf{z}})$ has been derived by Roscoe et al. (2013).

Here, the line-of-sight information alone is used to geometrically define a feasible range interval in order to simplify the process. Thus, the resulting region will have a rectangular shape. This is achieved by defining a smallest possible perigee $\|\mathbf{r}_{\min}\|$, a largest possible apogee radius $\|\mathbf{r}_{\max}\|$, and an allowed semi-major axis interval $[a_{\min}, a_{\max}]$ for all observations

$$\mathcal{C}(\mathbf{z}) = \{\rho_i, i \in [1,2] : \|\mathbf{r}_{\min}\| \leqslant \|\mathbf{r}(\rho_i)\| \leqslant \|\mathbf{r}_{\max}\|, a_{\min} \leqslant a(\rho_1,\rho_2) \leqslant a_{\max}\}. \tag{22}$$

Following the derivations in Schumacher et al. (2013) and Roscoe et al. (2013) the range bounds

$$\|\mathbf{r}_{\min}\| = a_{\min}(1 - e_{\max}) \tag{23}$$

and

$$\|\mathbf{r}_{\max}\| = a_{\max}(1 + e_{\max}), \tag{24}$$

are computed given the semi-major axis interval and the greatest allowed eccentricity $e_{\max}$.

As illustrated in Fig. 8, the radius for a given range ρ_i is given by

$$\|\mathbf{r}_i\|^2 = \rho_i^2 + \|\mathbf{r}_{s,i}\|^2 - 2\rho_i c_i, \tag{25}$$

where

$$c_i = (\mathbf{r}_{s,i} \cdot \mathbf{u}_i) \tag{26}$$

and i is either 1 or 2 for the first or second observation respectively. The bounding range interval for each tracklet is then computed with the following equations, where (25) has been rearranged for the range.

$$\rho_{\min,i} = -c_i + \sqrt{c_i^2 + \|\mathbf{r}_{\min}\|^2 - \|\mathbf{r}_{s,i}\|^2}, \tag{27}$$

$$\rho_{\max,i} = -c_i + \sqrt{c_i^2 + \|\mathbf{r}_{\max}\|^2 - \|\mathbf{r}_{s,i}\|^2}. \tag{28}$$

Additional to the range interval, the solutions to the Lambert's problem are also kept inside the semi-major axis bounds by defining an allowed interval of orbital revolutions

$$\begin{aligned} k_{\min} &= \lfloor 2\Delta t / P_(a_{\min}) \rfloor / 2 \\ k_{\max} &= \lfloor 2\Delta t / P_(a_{\max}) \rfloor / 2, \end{aligned} \tag{29}$$

using the orbital period P and the time interval between the two tracklets $\Delta t = t_2 - t_1$. The period is computed from the semi-major axis using Kepler's third law. The Lambert solver requires the number of completed full and half revolutions. Therefore, the values are rounded to the next lower 0.5 fraction.

5.2. Computational optimization

As mentioned above, the loss function $L(\mathbf{p}, k)$ must be minimized for all $k \in K = \{k_{\min}, k_{\min} + 0.5, \ldots, k_{\max}\}$

$$\mathbf{p}_{k,\min} = \arg\ \min_{\mathbf{p}\in\mathcal{C}} L(\mathbf{p}, k). \tag{30}$$

As the loss function is smooth enough, optimization algorithms can use the gradient information to efficiently reach the minimum. Prominent examples are conjugate gradient or quasi-Newton methods, where the latter has been used in this work.

The quasi-Newton methods approximate the loss function locally around some initial point $\mathbf{p}^* = (\rho_1^*, \rho_2^*)$ with a quadratic function. Then, the minimum is iteratively searched by finding the root of the gradient using Newton's method. If no analytical expression for the gradient is known, it can be provided using a finite difference scheme. Additionally, the inverse of the Hessian matrix of the loss function is required. Instead of computing it directly, it is approximated and iteratively improved during the minimization process. Different variants of the quasi-Newton method exists, resulting in different schemes on how to improve the matrix. An implementation for the popular Broyden–Fletcher–Goldfarb–Shanno (BGFS) scheme is used here and can be obtained from Press et al. (2007). For a detailed derivation of the algorithm, refer to Fletcher (1987).

As written above, the initial values $\mathbf{p}^*$ for the free parameters must be provided for each search iteration. That is why Kepler's third law is again used to determine a semi-major axis value $a(k)$ for each k using the time interval between the tracklets. The initial value for each range is then computed by

$$\rho_i^* = -c_i + \sqrt{c_i^2 + a(k)^2 - \|\mathbf{r}_{s,i}\|^2}, \tag{31}$$

assuming a circular orbit. This initial value is very close to the solution if the assumption is accurate.

After evaluating all k values, each one is selected that yields a loss function minimum below the predefined threshold. If at least one is obtained, the measurements are associated to each other with one orbit candidate. Multiple candidates are created if more k values fulfill the condition. These must be then validated with further measurements.

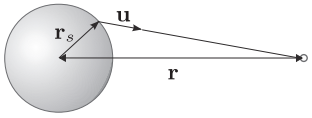


Fig. 8. Illustration of the measurement geometry.

6. Assessment and comparison

The proposed boundary-value method is now assessed in terms of association, robustness, and run-time performance. For this study, the optimization using the boundary-value formulation is first compared to the initial-value optimization. This comparison includes the effects on the robustness as well as on the run-time. Afterward, the computational burden is compared with other published methods, to illustrate the benefits of using the optimization when dealing with large data flows.

For that purpose, the Astronomical Institute of the University of Bern (AIUB) provided a set of measurements of the ZimSMART telescope (Zimmerwald SMall Aperture Robotic Telescope). The telescope scans the sky for objects in near-geostationary orbits. The scan is achieved in an automatic way, where declination stripes for a fixed right ascension value are captured. The declination range is beforehand determined considering the space object density of the region. A typical tracklet contains three to five single observations and its duration is between one and two minutes. Therefore, the spacing between individual

measurements of the tracklet is around 20 s. The characteristics of the measurement set are summarized in Table 2.

Herzog et al. (2013) assess the accuracy of the Zim-SMART astronomic positions using precise positions of navigation satellites. In addition to the noise (around 0.7,") they observe a systematic offset caused by systematic errors in the recording of the exposure epoch. For navigation satellites, along-track errors up to 10" are obtained. That is why the uncertainty attached to the measurements is described with two different components, i.e. the absolute accuracy and the relative one between individual observations within a tracklet (c.f. Siminski et al., 2013b). The latter must be used to determine the standard deviation of angular rates when performing a least squares fit, while the absolute accuracy, i.e. the spread of the systematic offset, only influences the modeled standard deviation of the angles.

The set was earlier processed by AIUB, i.e. the tracklets are tested with already cataloged objects. 180 tracklets could be successfully associated to previously observed objects. 40 of these objects have been observed two times in the measurement series, while 100 objects are only observed once. Although more tracklets are available, only the 180 tracklets that have been successfully associated are used for the performance test.

6.1. Association performance

Without any a priori information, i.e. in the build-up phase of a catalog or when a set consists of only uncorrelated tracklets, each tracklet must be tested against all others. This scenarios is referred to as an all-vs-all test in the following. In an optimal test, all tracklets that belong to the same object are associated to each other, while other tracklets are not grouped together.

The proposed optimization method for the boundary-value formulation is compared to the optimization method using the initial-value formulation. The association performance is tested by comparing the number of erroneous decisions for observations of non-common objects with the number of erroneous decisions for observations of common objects. The first is usually referred to as the false positive rate, while the latter describes the false negative associations. The all-vs-all tests are performed using different noise models in order to compare the robustness of the methods. The standard deviation of the assumed modeled noise is varied, i.e. $\sigma_{\alpha,\delta} \in \{0.7'', 1.2'', 2''\}$. This uncertainty is used to calculate the deviation of the angular rates from a least squares fit, which are then used in the loss function. The systematic error is assumed to be distributed with a standard deviation of 5".

Table 2
Specifications of the used measurements.

Specification	Value
First frame	Aug 18, 2012 22:59:08.64 UTC
Last frame	Aug 20, 2012 02:01:32.69 UTC
Total No. tracklets	204
No. identified tracklets	180
No. detected objects	140
No. re-observed objects	40
Tracklet duration	1–2 min
No. of measurements	3–5 per tracklet

The results of the test are shown in Fig. 9, where $\min L$ is computed for all tracklet pairs using the two formulations. The gating threshold is then altered to illustrate its effect on the respective association rates.

Both methods show similar results for the assumption $\sigma_{\alpha,\delta} = 0.7''$, in that most tracklets that belong to each other were successfully associated, while the false positive rate remains low. The initial-value formulation, however, behaves more sensitively to an increase in the noise value assumption than the boundary-value one.

In case of the initial-value formulation, a large uncertainty in the angular rates leads to even larger uncertainties in the modeled measurements. This ultimately leads to steeper ratio curves and more tracklet combinations being accepted as potential objects.

On the contrary, the impact of different models is not that dominant when using the boundary-value formulation. The angular rates uncertainties scale $\min L$, which shifts the curves slightly but the general shape remains constant. Thus, if no accurate error model is available, the boundary-value method allows for easier calibration.

Each accepted candidate, i.e. each hypothesis passing the decision threshold, needs to be taken into account for further measurement associations. A reasonable threshold must therefore be defined keeping the false positive rate low but still accepting enough associations to fill in the catalog. Further studies are necessary to define requirements for such a threshold.

6.2. Run-time performance

Another factor that needs to be assessed, especially when considering the large data flow of an optical sensor network, is the run-time performance. The all-vs-all test described above is used to estimate the computation time per tracklet pair. In total the number of association tests adds up to $\sim 16,000$.

Both optimization approaches, one using the boundary-value and one the initial-value formulation, have been implemented in C++ and are parallelized using OpenMP. The calculations for this work have been performed on an Intel Core i7–3720 CPU with 2.60 GHz clock rate.

For all association tests, the semi-major axis is forced to stay between 38,000 and 45,000 km, hence reducing the computational burden. The complete computation takes around 20 s for the boundary-value approach using a quasi-Newton method and 16 min for the initial-value method. The latter uses a differential evolution minimization scheme with 15 search points per generation and a maximum number of 50 generations (c.f. with method by

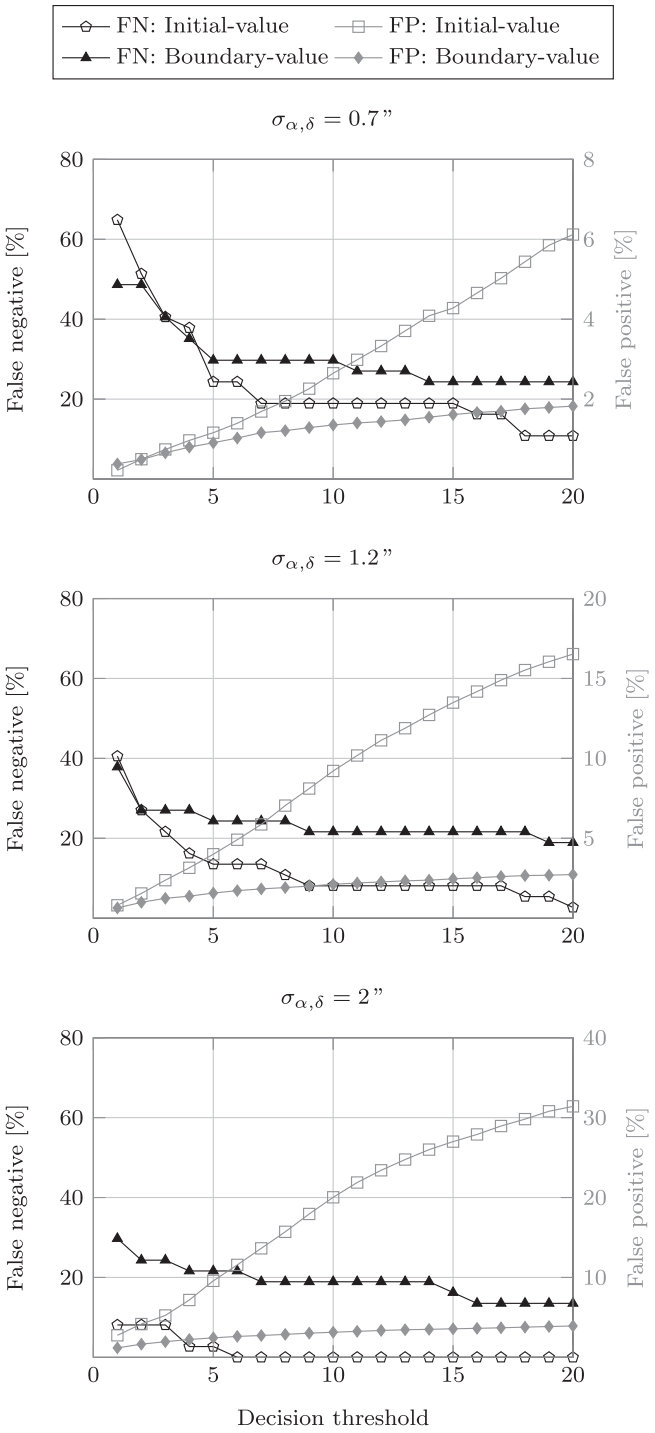


Fig. 9. False positive (FP) and false negative (FN) ratio dependent on the maximum allowed minimum loss function value $\min L$ (Decision threshold) for different measurement noise assumptions $\sigma_{\alpha,\delta} \in \{0.7'', 1.2'', 2''\}$.

Storn and Price, 1997). The test shows that the boundary-value formulation requires only $\sim 2\%$ of computational time compared to the initial-value optimization. One way to compare the run-time performance independent from the used computer is to assess the number of loss function calls. The number of function calls differs from each tracklet pairing test to the other as the topography of the loss function can be more or less challenging for the optimization algorithm. Therefore, the worst case scenario is discussed in the following paragraph.

With the given semi-major axis bounds, typically two discrete revolution cases need to be searched through for the provided data set using both methods. For a time separation of one day between the observations, the objects could have completed one full or a half revolution. When considering a separation of two days, the semi-major axis bounds allows only 1.5 or 2 completed revolutions. Thus, the maximum total number of loss function calls for the initial-value method sums up to around $1,500$, i.e. 2×50 generations $\times 15$ test points per generation. Similarly, the number of loss function calls for the boundary-value approach can be estimated. Typically each quasi-Newton search converges within 2–3 steps. A maximum number of steps has been defined to be 10 before the iteration is stopped. Therefore, even for the worst case, the total number of evaluations amount to 20, which is $\sim 1.33\%$ of the number of calls as when using the initial-value formulation. This theoretical result approximately resembles that of the previously mentioned computation time comparison. However, a perfect match is not obtained, as the cost of a loss function call differs between both formulations.

The run-time performance in terms of loss function calls can also be compared with methods proposed by other researchers. These comparisons must be interpreted with caution, as the actual computation time can be much smaller, e.g. using massive parallelization for the grid points. Furthermore, most grid based methods can be accelerated by allowing a courser discretization, which typically increases the risk of missed associations. Still, when dealing with a lot of measurements and limited resources, it gives a good estimate of the computational burden.

Fujimoto et al. (2013) analyzes a similar data set and uses 750×500 hypothesis bins for each pairing test. Consequently, for each constellation this number of hypotheses must be propagated to the common epoch, i.e. the epoch of one observation. Neglecting the additional computations when comparing each hypothesis with all bins of the other observation arc, the overall burden is similar to $375,000$ function evaluations. Gadaleta et al. (2012) uses 150×100 grid points to sample a feasible partition of the admissible region for simulated geostationary objects. All hypotheses are then individually propagated and compared to the next observation using different testing metrics, which leads to a computational burden similar to $15,000$ function evaluations.

6.3. Discussion

The findings of the previous section are now discussed. Advantages and difficulties are compared for the boundary-value and initial-value formulation.

The initial-value method can incorporate any kind of orbit propagator and is not restricted to a purely Keplerian orbit model. It is well suited for use with numerical

trajectory integration and enables use of fidelity force models. This is of particular interest for high-area-to-mass ratio (HAMR) objects affected by large non-gravitational perturbations. However, this would also add the solar radiation pressure coefficient as a free parameter to the optimization problem.

A major drawback is that the method is very sensitive to false noise model assumptions as can be observed in Section 6.1. Furthermore, as the loss function is dependent on both angles and angular rates, a false noise model might lead to a different shape that favors either angles or the rates. Then, in addition to poor association performance, the obtained orbit solution might be also largely off the target.

The loss function for the initial-value formulation so far does not allow focusing on one valley at a time, i.e. the individual local minima cannot be separated easily. Consequently, it cannot be guaranteed to find the global minimum. Additionally, the computation time is increased as the admissible region must be sampled with a large number of initial start points for each search (c.f. with results in Section 6.2).

The boundary-value formulation is a promising approach: the simple shape of the loss function enables the use of gradient-based optimization methods which speeds up the computation. Furthermore, as the different local solutions are already separated by the Lambert solver itself, it is generally more stable than the initial-value method. The global minimum inside the admissible region is definitely found. The separation also speeds up the computation as only one start point for each search iteration is used. The total number of loss function calls outperforms all other presented methods.

The used loss function relies solely on the angular rates. That makes it more tolerant against a false noise model as illustrated in Section 6.1.

Traditional solutions to the Lambert's problem typically only account for two-body dynamics. This can cause difficulties when associating measurements that are separated by long time spans or objects with dominant non-conservative forces. However, this difficulty can be circumvented by using numerical orbit models and modern shooting methods for the Lambert's problem that also account for perturbations.

7. Summary and conclusion

An advanced method for the association of tracklet pairs has been presented. The approach solves the tracklet association and initial orbit determination using a boundary-value formulation. It finds the best matching range pair given two tracklets at different observation epochs. It uses the solution to the Lambert's problem to calculate orbit candidates and the angular rates to distinguish between good and bad candidates. The initial orbit determination is reformulated into an optimization problem. The shape of the minimized loss function is very suitable for optimizers, which ultimately leads to a rapidly converging association and orbit determination process.

The approach promises to be a robust and fast replacement for current methods used in space object cataloging tasks. To confirm the suitability, an evaluation and comparison of the different approaches has been performed using a data set from the ZimSMART telescope. The results are summarized as follows: First, current methods using an initial-value formulation are very computationally demanding. More importantly, they require very accurate knowledge on the errors and noise and are thus not very robust to false assumptions. Second, the loss function based on initial-value formulations is difficult to minimize as multiple local minima must be identified. This study has shown that the presented approach can overcome this weaknesses. The optimization algorithm using the boundary-value formulation will definitely find the global minimum in the specified admissible region. Third, the robustness w.r.t. to false noise model assumptions has been demonstrated. Lastly, the presented approach requires much less computational resources than other competing methods.

So far, the boundary value method is initialized assuming the objects from the geostationary domain. However, the method can be extended to other orbit types, e.g. highly eccentric ones like Molniya orbits. It is recommended that further research be undertaken in the assessment of noise and error models. These are required to define reasonable decision thresholds, which ascertain whether object candidates are created or not. Further investigations are also needed to determine which threshold and therefore false positive and false negative ratio is beneficial for an efficient but still all-encompassing catalog.

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